Laboratory Record

of _____

Roll No.: _____
Experiment No.: _____
Sheet No.: _____
Date.: _____

Aim

To build a dynamic web application that calculates a student's Semester Grade Point Average (SGPA) and updated Cumulative Grade Point Average (CGPA).

Description

A web-based tool where users input their previous academic data and current semester performance. JavaScript dynamically generates the required number of subject input fields, maps letter grades to numeric values, and calculates both the semester's SGPA and the updated overall CGPA.

Algorithm

1. Setup: Map letter grades (S, A, B, etc.) to their respective numeric points (10, 9, 8, etc.).
2. Generate Fields: On clicking "Generate", dynamically create dropdowns for grades and inputs for credits based on the specified number of subjects.
3. Read Data: On clicking "Calculate", fetch the previous CGPA, previous credits, and the newly entered grades and credits for the current semester.
4. Compute SGPA: Calculate the total current semester points (grade points × credits) and divide by the total current semester credits.
5. Compute CGPA: Combine the previous total points (Previous CGPA × Previous Credits) with the current semester points, then divide by the overall total credits.
6. Output: Display the calculated SGPA and new CGPA on the screen.

Program Code

Index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>SGPA & CGPA Calculator</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <div class="calculator">
    <h2>SGPA & CGPA Calculator</h2>
    <input type="number" id="prevCGPA" placeholder="Previous CGPA">
```

```
    <input type="number" id="prevCredits" placeholder="Previous Credits">
    <br><br>
    <input type="number" id="subjectCount" placeholder="Number of Subjects">
    <button onclick="generateSubjects()">Generate</button>
    <div id="subjects"></div>
    <button onclick="calculate()">Calculate</button>
    <div id="result"></div>
  </div>
  <script src="script.js"></script>
</body>
</html>
```

Script.js

```
const gradePoints = {
  S: 10,
  A: 9,
  B: 8,
  C: 7,
  D: 6,
  E: 5,
  F: 0
};
function generateSubjects() {
  const count = Number(document.getElementById("subjectCount").value);
  const subjects = document.getElementById("subjects");
  subjects.innerHTML = "";
  for (let i = 1; i <= count; i++) {
    subjects.innerHTML += `
      <div>
        Subject ${i}
```

```
            <select class="grade">
                <option value="">Grade</option>
                <option value="S">S</option>
                <option value="A">A</option>
                <option value="B">B</option>
                <option value="C">C</option>
                <option value="D">D</option>
                <option value="E">E</option>
                <option value="F">F</option>
            </select>
            <input type="number" class="credit" placeholder="Credits">
        </div>
    `;
  }
}
function calculate() {
  const prevCGPA = Number(document.getElementById("prevCGPA").value);
  const prevCredits = Number(document.getElementById("prevCredits").value);
  const grades = document.querySelectorAll(".grade");
  const credits = document.querySelectorAll(".credit");
  let totalPoints = 0;
  let totalCredits = 0;
  for (let i = 0; i < grades.length; i++) {
    const grade = grades[i].value;
    const credit = Number(credits[i].value);
    if (grade && credit) {
      totalPoints += gradePoints[grade] * credit;
      totalCredits += credit;
    }
```

```
    }
    const sgpa = totalCredits > 0 ? (totalPoints / totalCredits) : 0;
    const totalOverallCredits = prevCredits + totalCredits;
    const newCGPA = totalOverallCredits > 0
        ? ((prevCGPA * prevCredits) + (sgpa * totalCredits)) / totalOverallCredits
        : 0;
    document.getElementById("result").innerHTML = `
        <p>SGPA: ${sgpa.toFixed(2)}</p>
        <p>New CGPA: ${newCGPA.toFixed(2)}</p>
    `;
}
```

Style.css

```
body {
    font-family: Arial, sans-serif;
    background: #f5f5f5;
    padding: 30px;
}
.calculator {
    width: 400px;
    margin: auto;
    background: white;
    padding: 20px;
    border: 1px solid #ddd;
    border-radius: 8px;
}
h2 {
    text-align: center;
}
input,
```

```
select,
button {
    padding: 8px;
    margin: 5px;
}
#subjects div {
    margin: 10px 0;
}
button {
    cursor: pointer;
    border: 1px solid #ccc;
    background: white;
    border-radius: 4px;
}
button:hover {
    background: #eee;
}
#result {
    margin-top: 20px;
    font-weight: bold;
    text-align: center;
}
```

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SGPA & CGPA Calculator

9.09 | 48

6 | Generate

Subject 1	S	3
Subject 2	B	2
Subject 3	A	4
Subject 4	S	2
Subject 5	A	1
Subject 6	C	0

Calculate

SGPA: 9.25

New CGPA: 9.12